CS2201 Computer Organization & Architecture

L34: Introduction to PCIe and USB standards



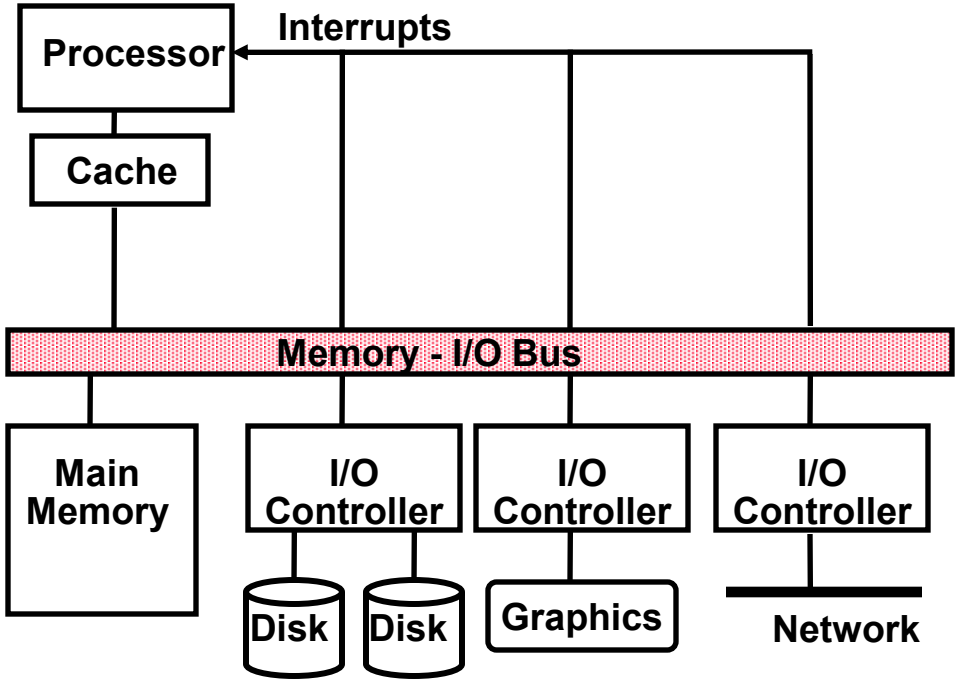
John Jose

Professor

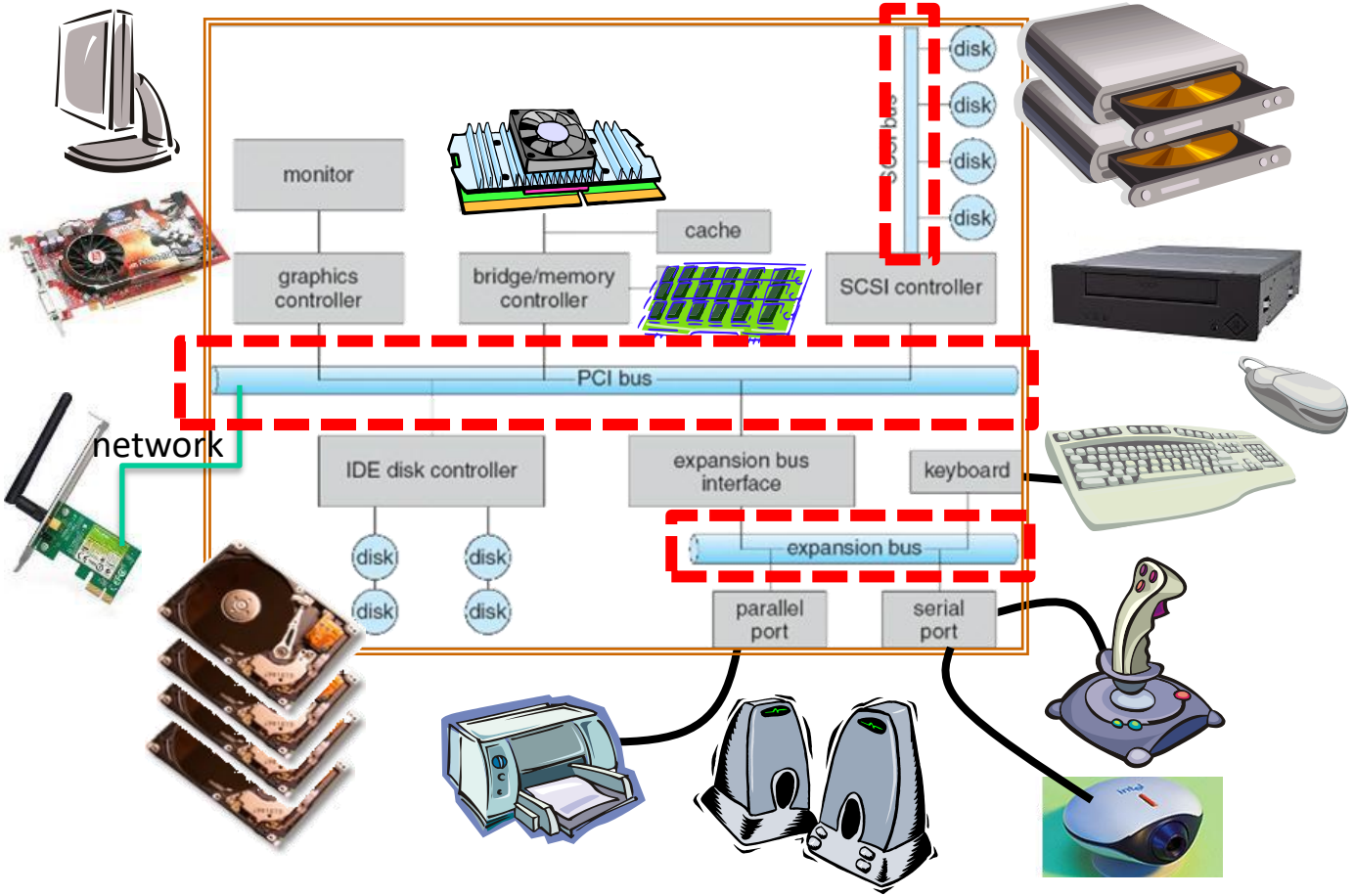
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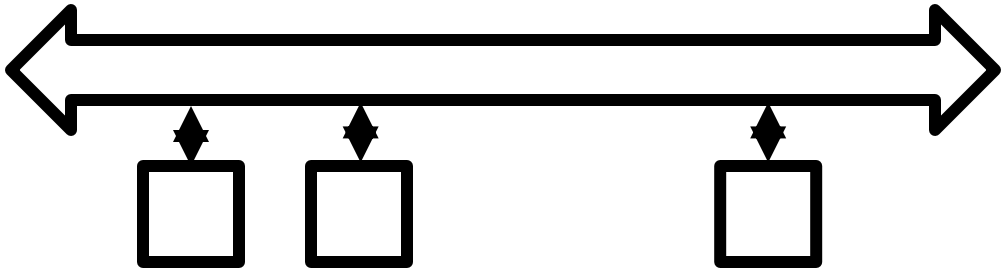
Typical I/O Subsystem



Example of I/O System

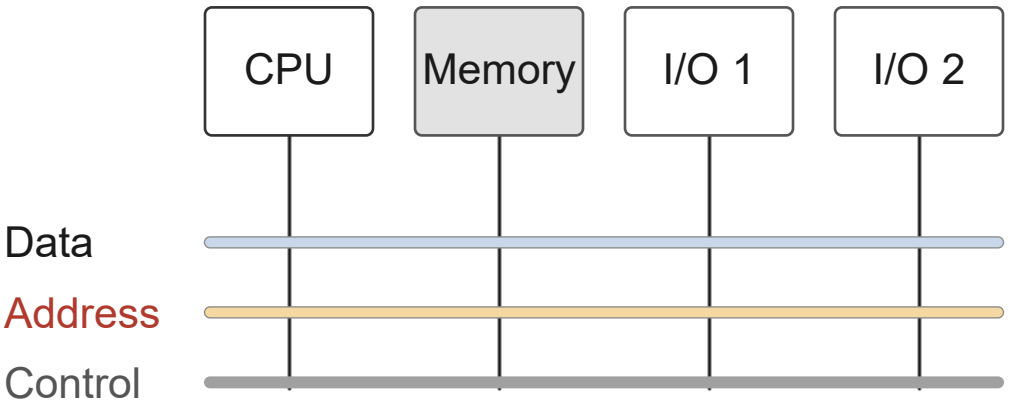


What's a bus?



- Common set of wires for communication among hardware devices plus protocols for carrying out data transfer transactions
  - Operations: e.g., Read, Write
  - Control lines, Address lines, Data lines
  - Typically multiple devices
- Protocol: initiator requests access, arbitration to grant, identification of recipient, handshake to convey address, length, data

System Bus



Data lines

Width = bits moved per transfer

Address lines

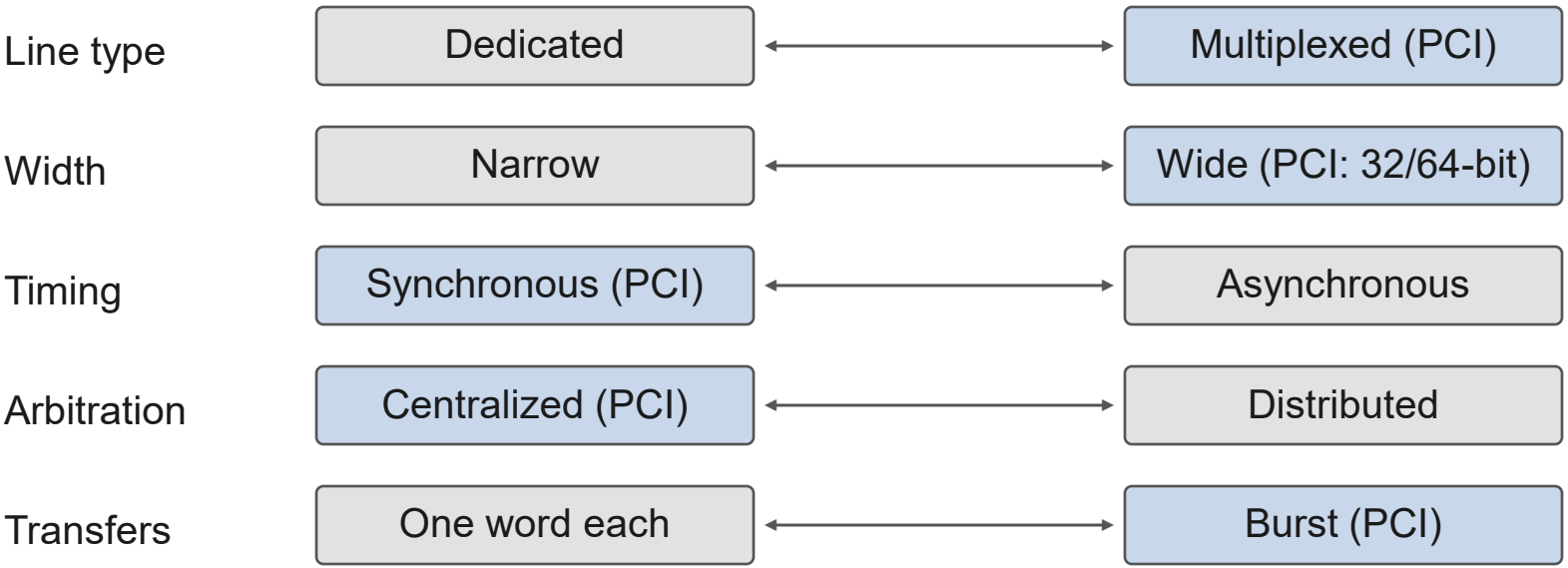
Width = size of the addressable space

Control lines

Read/write, clock, request/grant, interrupt, reset

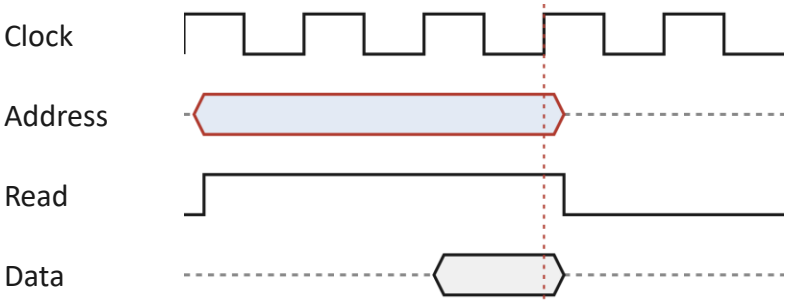
Bandwidth = width × transfer rate
64 bits × 100 M transfers/s = 800 MB/s

Bus Design Elements

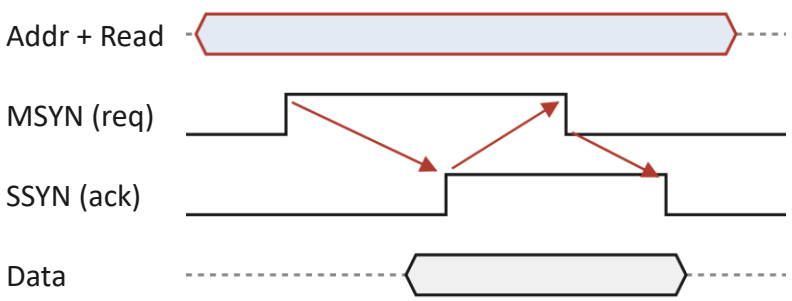


Bus Timing

Synchronous read: everything on clock edges

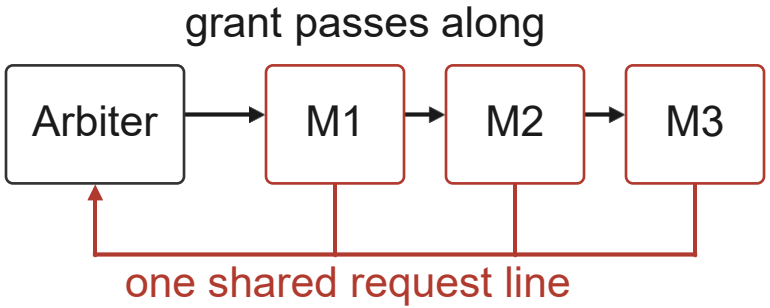


Asynchronous read: a four-step handshake



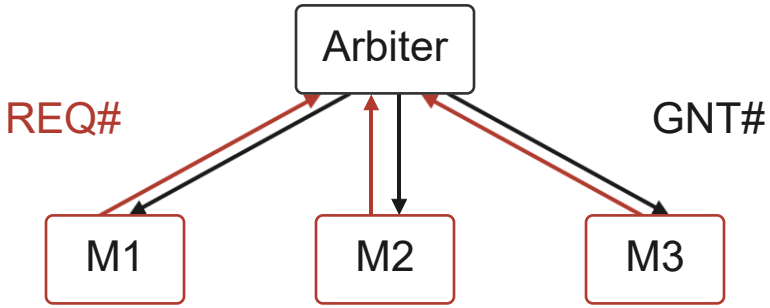
Bus Arbitration

Daisy-chained grant



Few wires, but position = priority

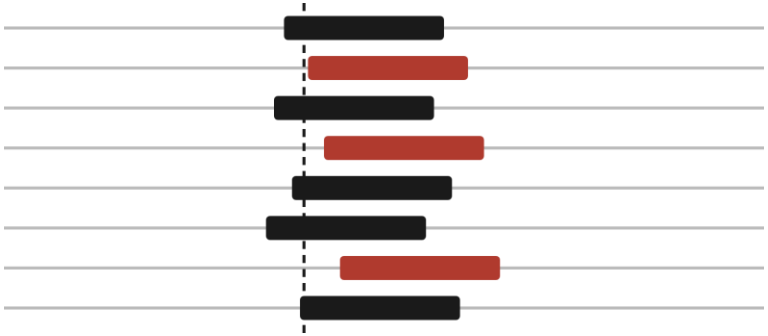
Independent request / grant



Fast, any policy, PCI uses this

Parallel vs Serial

Parallel: 8 bits sent together…



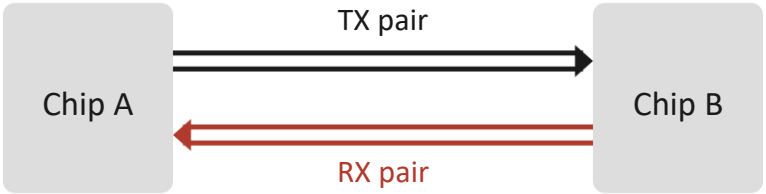
…but arrive at different times (skew)

each slot adds load

one talker at a time

many pins

Serial: point-to-point lanes



clock embedded in data

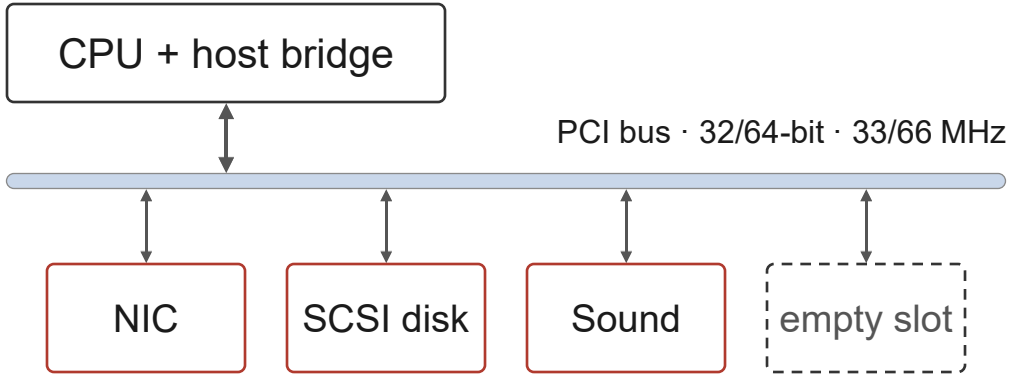
1 driver, 1 receiver

full duplex

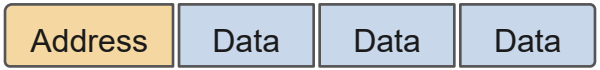
add lanes: ×1 … ×16

PCI

PCI, introduced by Intel in 1992 and then managed by the PCI-SIG



Same wires carry address, then data



one address, then a burst

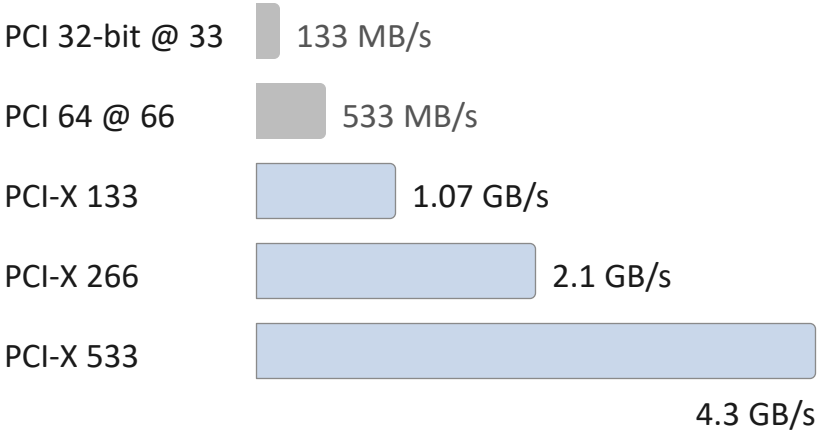
133 MB/s

= 4 bytes × 33 MHz

64-bit at 66 MHz → 533 MB/s

PCI-X

Peak bandwidth

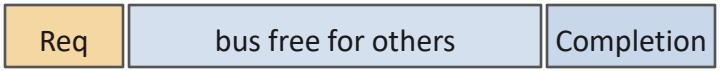


Split transactions

PCI: a slow target makes the master retry



PCI-X: target lets go of the bus, replies later



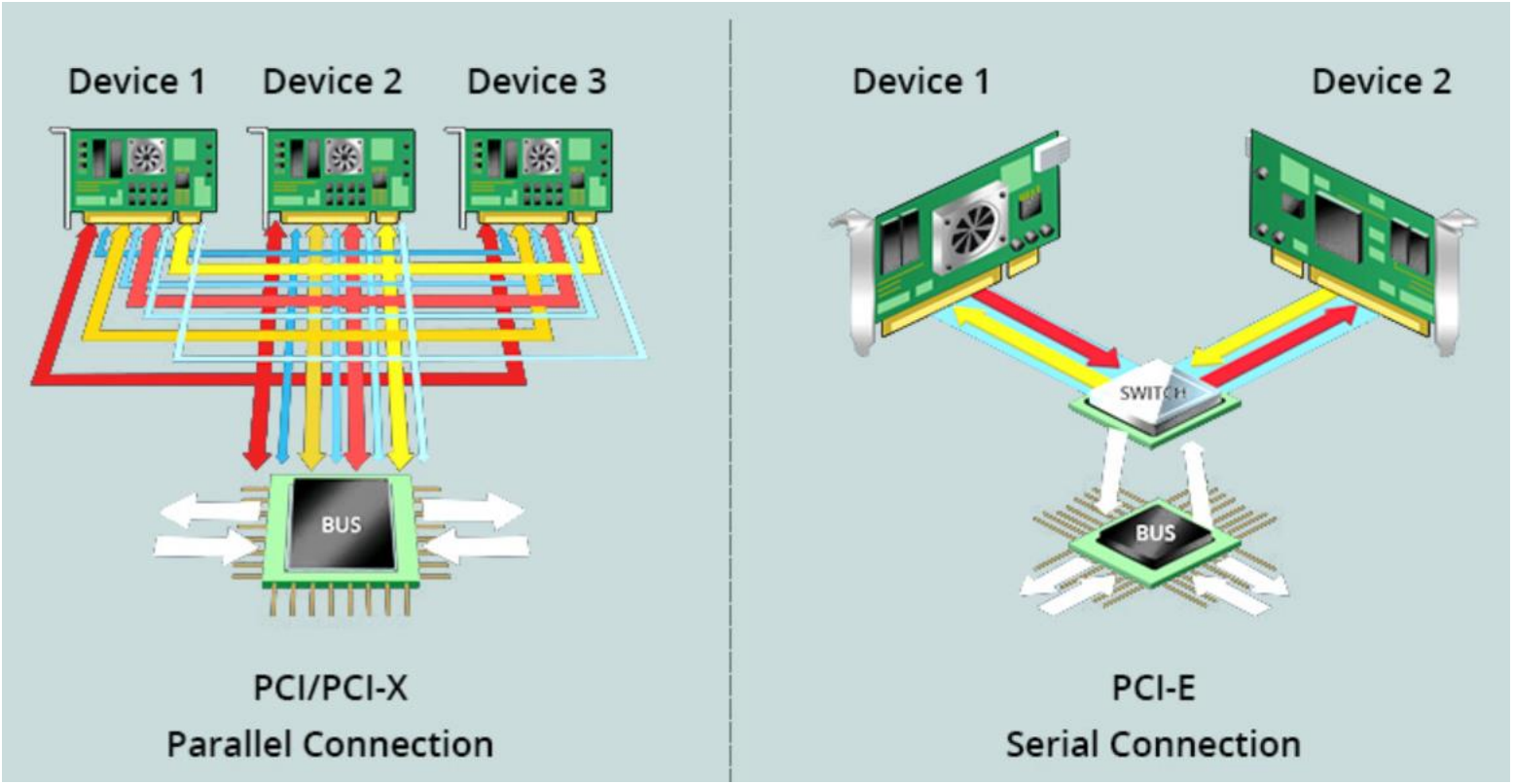
PCI Vs PCI-X

Feature	Conventional PCI	PCI-X (PCI eXtended)
Primary Target Market	**Consumer desktop computers**	**Enterprise servers and workstations**
Year Introduced	1992 (by Intel)	1998 (by IBM, HP, and Compaq)
Bus Type	Parallel (shared topology)	Parallel (shared topology)
Bus Width	32-bit (standard) or 64-bit	64-bit
Clock Speed	33 MHz or 66 MHz	Up to 133 MHz (v1.0) / up to 533 MHz (v2.0)
Maximum Bandwidth	133 MB/s (at 32-bit/33 MHz)	1.06 GB/s (v1.0) / 4.26 GB/s (v2.0)

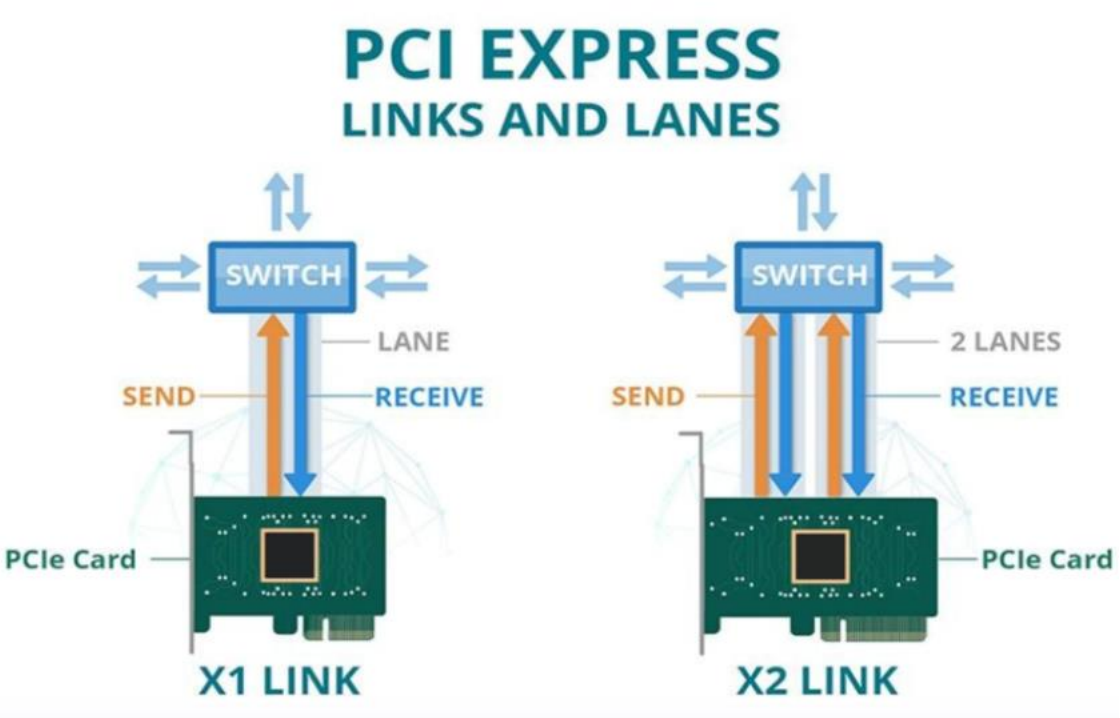
PCI-X Vs PCI-e

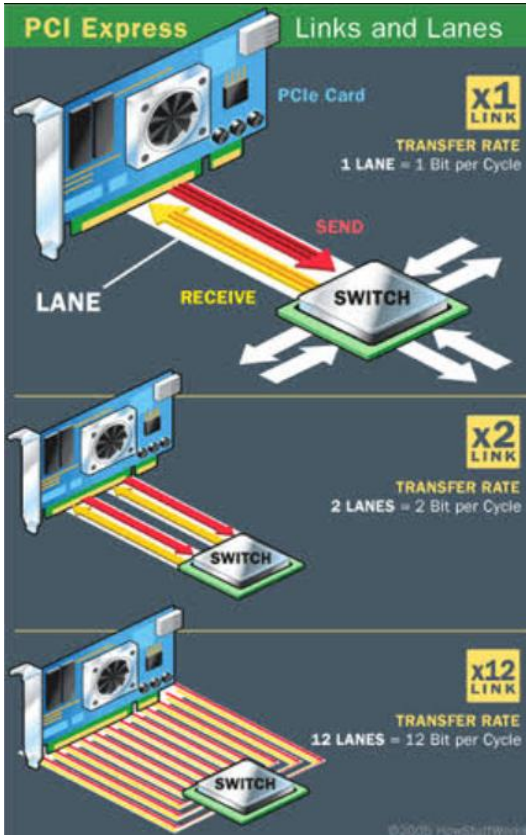
Feature	PCI-X (PCI Extended)	PCI-e (PCI Express)
Architecture	**Parallel bus:** An extension of conventional 64-bit PCI.	**Serial bus:** Completely redesigned packet-based architecture.
Topology	**Shared:** All devices on the bus segment share the same bandwidth.	**Point-to-Point:** Every device has its own dedicated connection.
Data Flow	**Half-Duplex:** Can only transmit data in one direction at a time.	**Full-Duplex:** Simultaneously sends and receives data.
Max Throughput	Peaked at **1.06 GB/s** (PCI-X 2.0 / 133MHz).	Scalable from **4 GB/s up to 256 GB/s+** (Generations 4.0 through 7.0).
Primary Use Case	Legacy enterprise servers and workstations (now obsolete).	Universal standard for all modern PCs, servers, and GPUs.

PCI-X Vs PCI-e


Device 1
Device 2
Device 3
BUS
PCI/PCI-X
Parallel Connection
Device 1
Device 2
SWITCH
BUS
PCI-E
Serial Connection

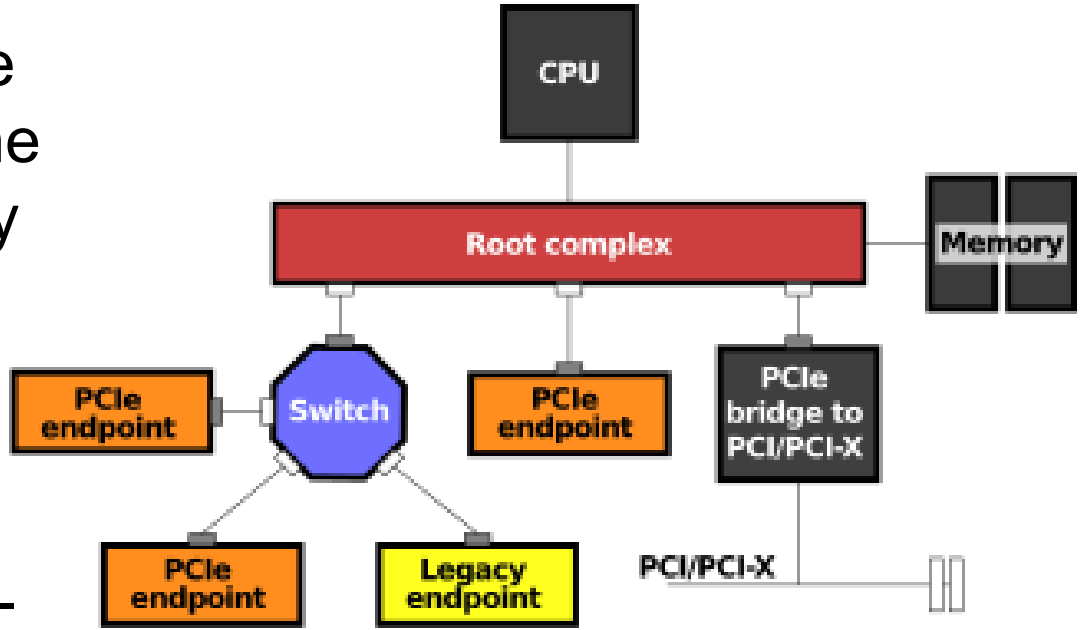
PCI-e Links


PCI EXPRESS
LINKS AND LANES
SWITCH
LANE
SEND
RECEIVE
PCIe Card
X1 LINK
SWITCH
2 LANES
SEND
RECEIVE
PCIe Card
X2 LINK


PCI Express
Links and Lanes
PCIe Card
x1 LINK
TRANSFER RATE
1 LANE = 1 Bit per Cycle
SEND
LANE
RECEIVE
SWITCH
x2 LINK
TRANSFER RATE
2 LANES = 2 Bit per Cycle
SWITCH
x12 LINK
TRANSFER RATE
12 LANES = 12 Bit per Cycle
SWITCH

System Diagram with PCI-e

- **PCI Express Root Complex** is the hardware interface that connects the CPU and system memory to the rest PCIe I/O hierarchy
- Translating and routing data between the processor/RAM and high-speed peripherals



USB Architecture

- **USB - Universal Serial Bus**.
- **Industry-standard specification** for cables, connectors, and communication protocols used to connect, communicate, and supply electric power between personal computers, mobile devices, and peripheral accessories
- **Host-to-device (master-slave) architecture** – central host (computer/tablet/smartphone) controls all communication on the bus, and peripheral devices respond only when instructed by the host.

USB Architecture

- **Detection:** Change in voltage signaling at host →a new device
- **Enumeration:** The host sends a request to the device to identify itself, asking for its vendor ID, device type, and power requirements.
- **Address Assignment:** The host assigns the device a unique identification address
- **Driver Loading:** The operating system installs the appropriate device driver so the user can immediately use it without restarting the computer (plug-and-play).

USB Architecture

- Data Transmission - Data is broken down into small packets not sent as a continuous stream.
- Different transfer types:
  - **Control:** For configuration commands.
  - **Bulk:** For large data blocks (like moving a file to a flash drive or printing a document).
  - **Interrupt:** For small, time-sensitive data (like mouse movements or keystrokes).
  - **Isochronous:** For continuous, real-time data where timing is critical (like audio streams or live video)

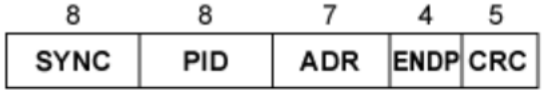
USB transfer types

A USB packet is the basic, individual building block of data transmission on a USB bus, while a USB frame is a fixed-time window managed by the host that organizes these packets and transactions over time.

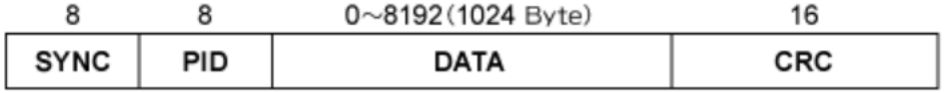


Packet Type	PID Value	Packet Identifier
Token	0001	OUT Token
	1001	IN Token
	0101	SOF Token
	1101	SETUP Token
Data	0011	DATA0
	1011	DATA1
	0111	DATA2
	1111	MDATA
Handshake	0010	ACK
	1010	NAK
	1110	STALL
	0110	NYET
Special	1100	Preamble
	1100	ERR
	1000	Split
	0100	Ping

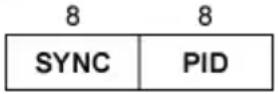
Token Packet



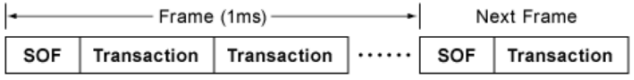
Data Packet



Handshake Packet

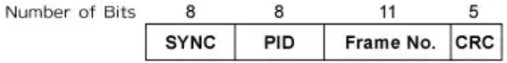


Frame Format



one frame: 1 ms (full speed) · 125 µs microframe (high speed)

SOF Packet





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